

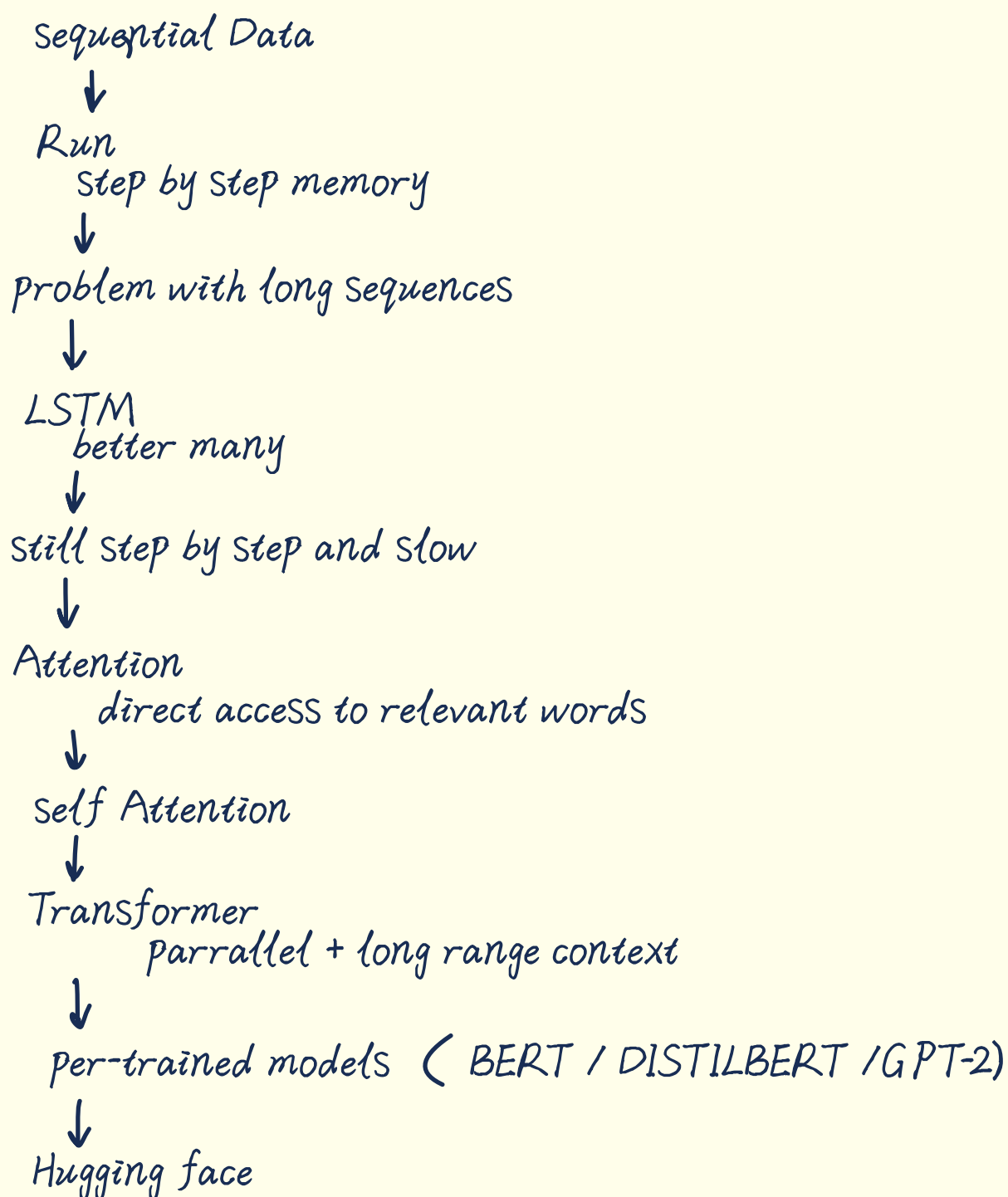
وَقُلْ رَبِّ زِدْنِي عِلْمًا

Day 4 Attention & Transformers

objectives:

- 1- Limitation of RNN/LSTM
- 2- Attention
- 3- Self-Attention
- 4- Parallelism
- 5- transformer Architecture
- 6- Positional Encoding
- 7- pre-trained Transformers
- 8- Hugging face

So the goal is



Attention

Suppose we have a Statement

Sara put the book on the table because it was heavy

when the model reaches to (it) the Model should know it refers to book or table

≡ The RNN/LSTM was transmitting the information step by step

Sara → put → the → book → . . . → it

But Attention is different → Instead of finding the current word through long series of steps, the current word is estimated by looking directly at all the previous words and deciding which word is more important to it

for Example

Sara → importance (Low)

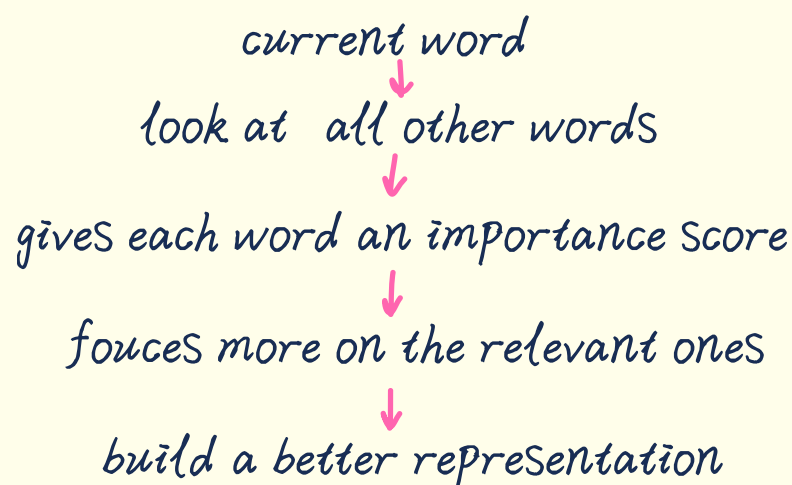
book → importance (high)

table → importance (medium)

heavy → context

So Attention focus more on the most related word

Attention workflow



the attention is important because it helps in three things

- Self Attention

- Parallelism

- Long Range context

So the attention lets each word directly focus on the other words that are most relevant to it, instead of relying only on step-by-step memory

Self Attention

Self Attention means that each word within the same sentence looks at the rest of the words and determines which are more important to understand its meaning or

Self Attention allows each word to look at all other words in the same sequence and decide which ones are most relevant

For example

the movie was long, but it was interesting

when the model processes (it) → it needs to know that (it) is linked to (movie)
so self attention allows (it) to look directly at the (movie) instead of relying only on memory transfers step by step

Query, key, value

Each word represented by three things

- Query (what looking for)
- key (what information)
- value (what information should be given)

For example

the movie was long, but it was interesting

***** use the Query that follows it and compare it with the key for the rest of the words
if movie related with it, it gets High Attention Score, then the model takes more information than the value following the movie

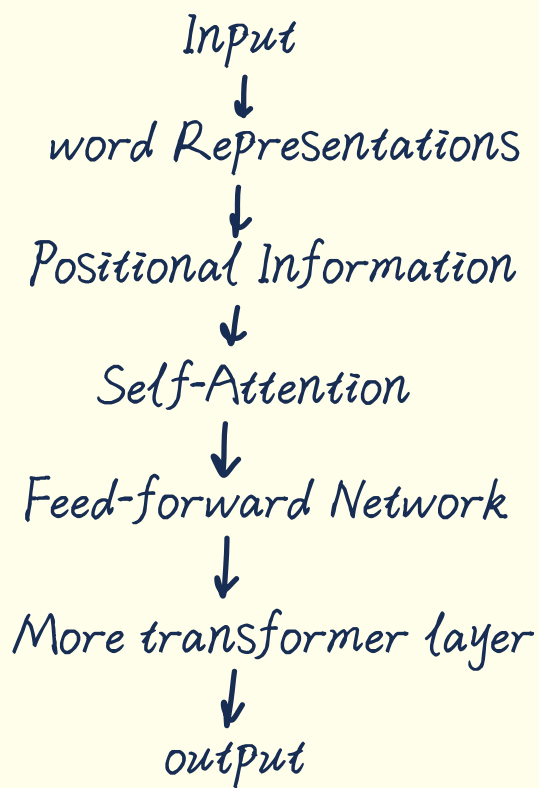
word 1 ↔ word 2 ↔ word 3 ↔ word 4

every word can look directly at other words because of this self attention has 2 features

- 1- long range context :- The word can associate itself with a distant word directly
- 2- Do not need to wait to process one word, then the second, then the third
(multi sites can be processed together)

Transformer

Transformer is an architecture essentially built around (self attention) instead of the recurrence found in RNN/LSTM



problem

if transformer processes words together how know the order?

For example

- Dog bites man
- Man bites Dog

same words diff's order & the Transformer doesn't have a recurrence that says

First → Second → third

So we need **positional Encoding**

positional Encoding

word meaning
+
word position

positional encoding gives the transformers information about the order of words

Add information to each word that tell the Model

I am word 1
I am word 2
I am word 3
:
:

Multi-Head Attention

Multi Head Attention allows the model to learn different relationships between words at the same time

why? because each head can learn a different relationship

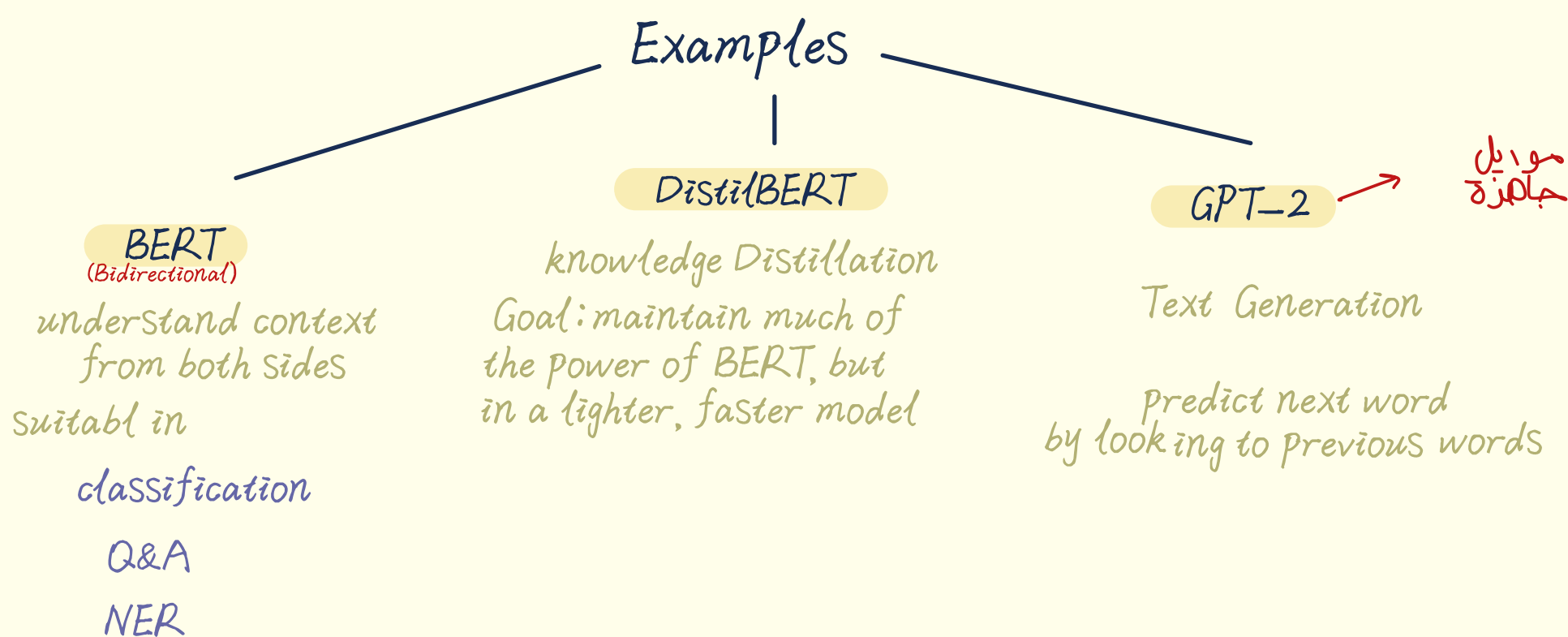
For Example : The movie that save matched was excellent

- Head 1 → Focus on relationship between movie ↔ excellent
Head 2 → Focus on relationship between Sara ↔ watched
Head 3 → Focus on relationship between watched ↔ movie

pre-trained Transformers

Instead of getting a new transformer and teaching it the language from scratch, we get a model that was previously learned from it, and then we use it on modify it for a task such as: sentiment analysis, text classification

So pre-trained transformer = Transformer already trained on a large amount of text, which we reuse for another NLP task



Hugging face

It provides libraries and tools that make it easier to download ready-made models and use them instead of building everything manually

RNN / LSTM VS Transformer

RNN and LSTM: process sequence, step by step using recurrent memory

while transformer : use Self-Attention to directly connect different positions in the sequence and allow much more parallel processing

Feature	RNN	LSTM	Transformer
Processing	Step by step	Step by step	Processes positions more in parallel
Memory	Hidden state	Hidden state + cell state	Self-attention
Long-range Dependencies	Difficult	Better than RNN	Strong direct access across the sequence
Main Mechanism	Recurrence	Recurrence + gates	Self-attention
Parallel Processing	Limited	Limited	Much better
Long Sequences	Can struggle	Handles them better	Better suited to long-range relationships
Training Speed	Relatively slow	Relatively slow	More parallelizable
Typical NLP Use	Basic sequence modeling	Sequence/text modeling	Modern NLP models such as BERT and GPT

